

Day 6

Layout algorithms

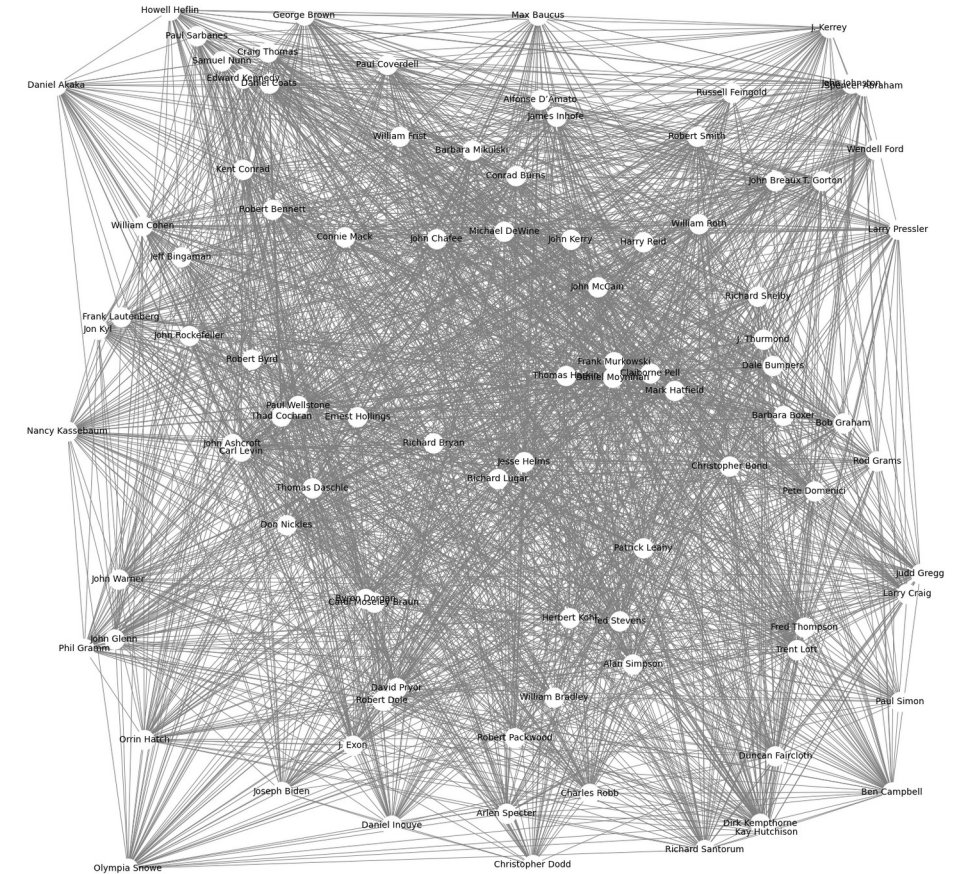
Lecturer: Msc. Minh Tan Le

Outlines

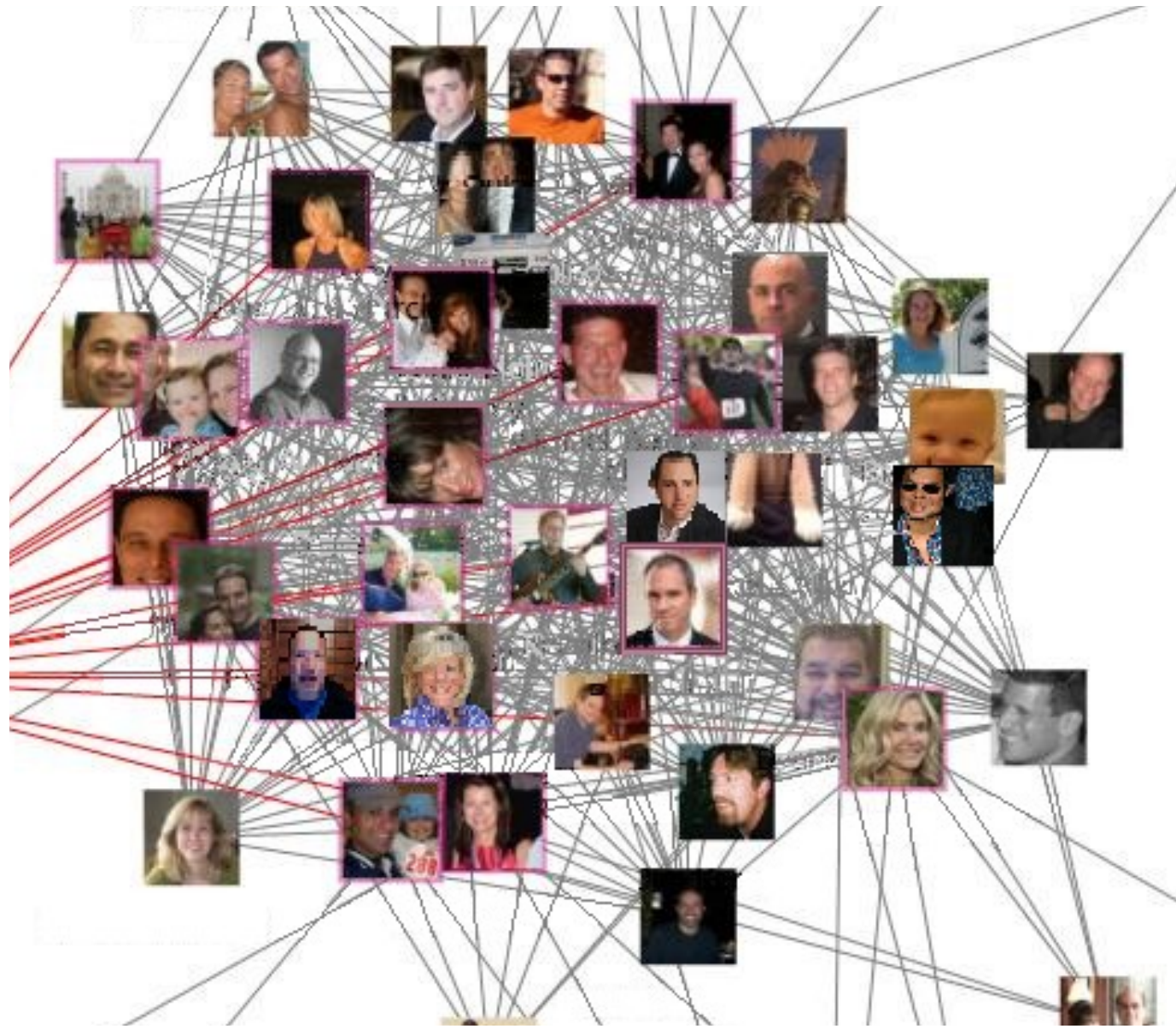
- I. Easy network layouts
- II. Advanced layouts
 - Physics-driven layouts
 - Spectral
 - Kamada Kawai
 - Fruchterman – Reingold
- III. Data-driven layout

I. Easy network layouts

- Random, circular & spiral.
- Each have their own usability.

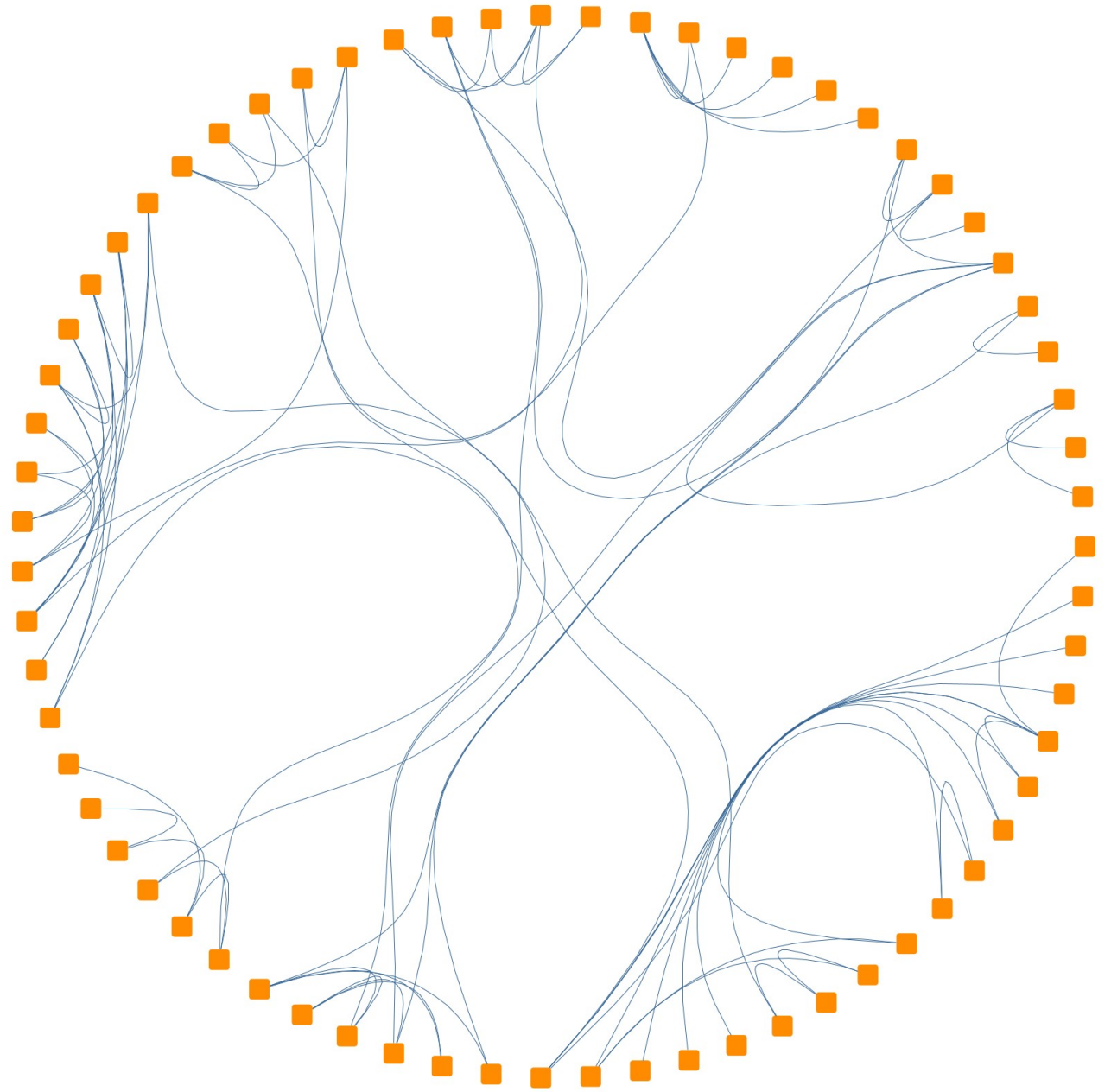


*Random layout
For density illustration*



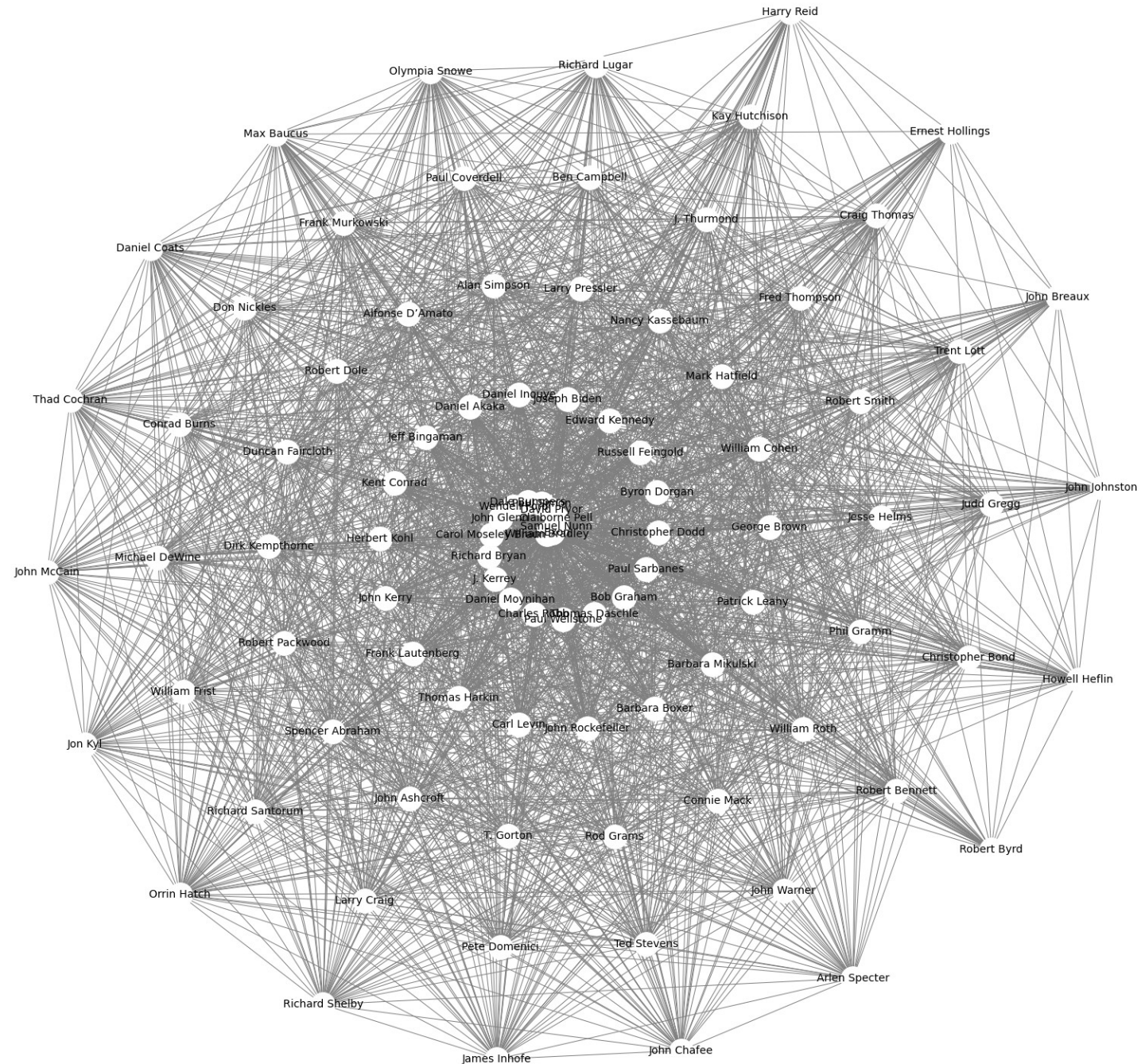
Marketing

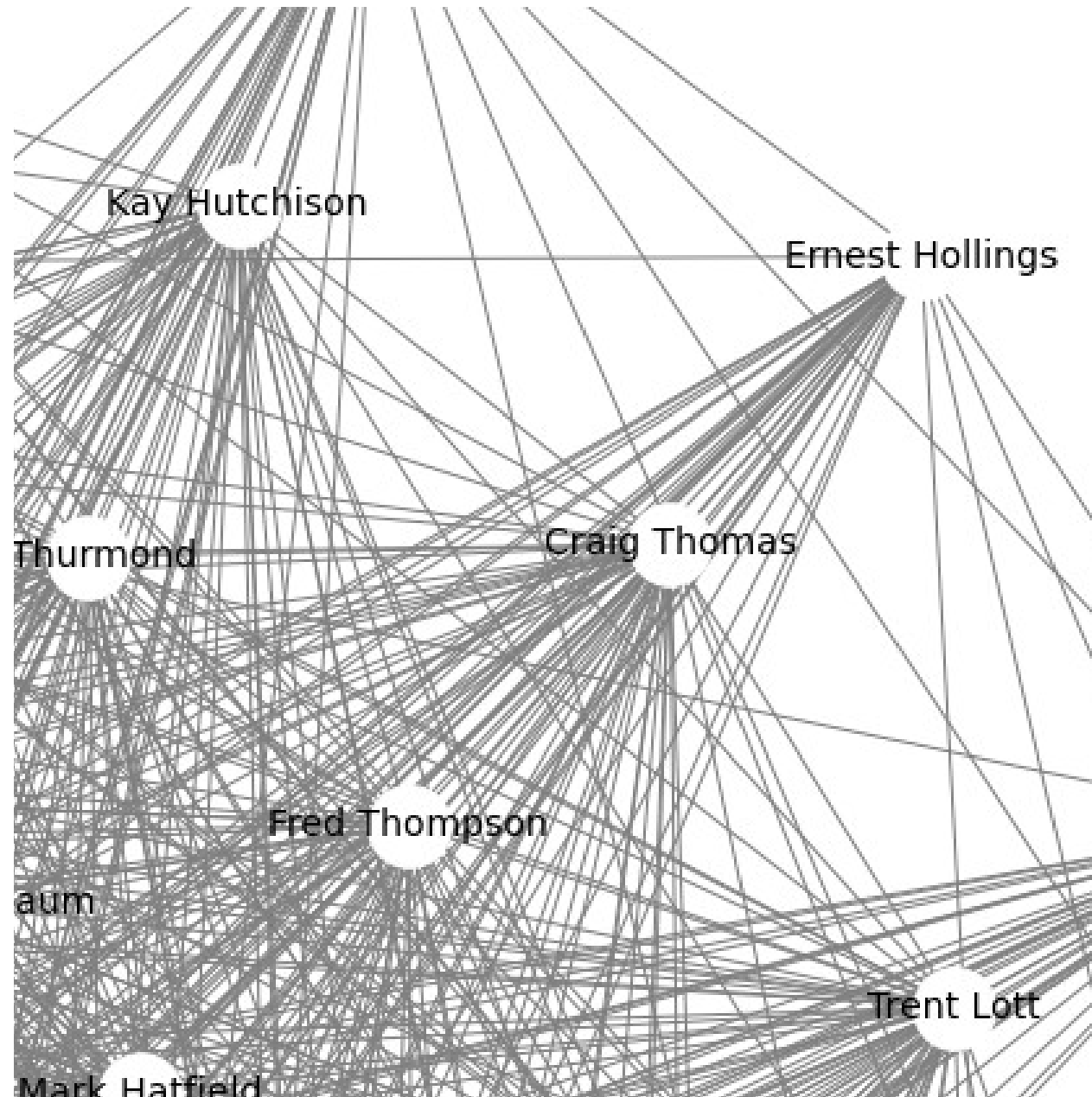
Circular: Use when there are multiple clusters with limited cluster-to-cluster connections.
Rule: Nodes in same group should be close together.



Spiral: Same as circular's purpose, but with many connections.

Rule: Less-connected points are far away from center.



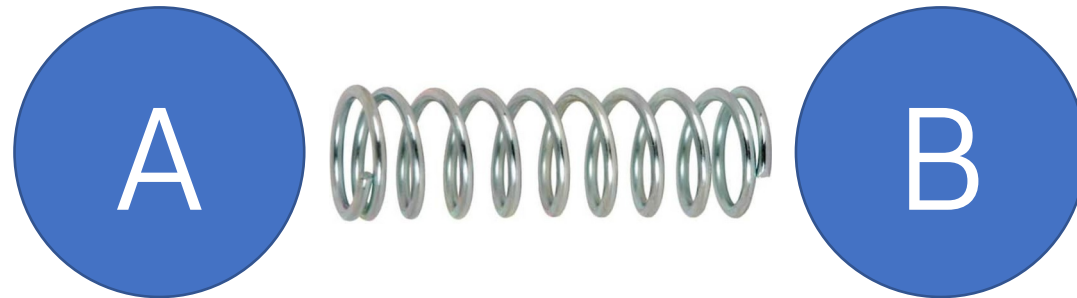


II. Advanced layouts

- Group of layouts that positioning nodes by connections.
- Purposes: Improve readability, increase interaction.
 - Varied edge lengths.
 - Lessen crossed links.
- Drawback: Can be slow with huge data.

Spring system (Force-directed graph drawing):

- There is energy (ε) in each edge.
- Springs should be balanced.
- Spring can push out or drag into.



1. Spectral

- Ideas:
 - Those nodes with many connections should stick with low degree ones.
 - The connected nodes are linked by a spring.
 - Deterministic: The node positions are unchanged after multiple runs.
 - Optimistic: Acceptable time complexity.

$$G = (V, E), |V| = m$$

$$g(G) = H = (W, F), |W| = n$$

$$n \leq m$$

$$\rho: V \rightarrow \mathbb{R}^n$$

$$P \in R^{m \times n}$$

$$n \leq m$$

$$\rho(v_0)$$

$$\rho(v_1)$$

$$\rho(v_2)$$

$$\rho(v_3)$$

 m [illegible]

$$P \in R^{m \times n}$$

$$n \leq m$$

$$\varepsilon(P) = \sum_{\{v_i, v_j\} \in E} \|\rho(v_i) - \rho(v_j)\|^2$$

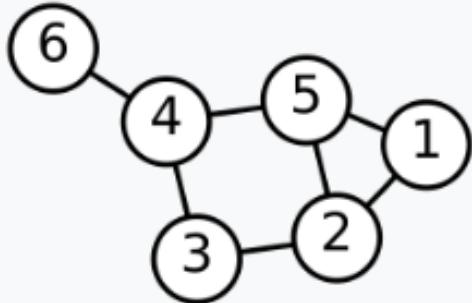
$$P \in R^{m \times n}$$

$$n \leq m$$

$$\varepsilon(P) = \sum_{\{v_i, v_j\} \in E} \|\rho(v_i) - \rho(v_j)\|^2$$

$$L_{i,j} = \begin{cases} \deg(v_i) & \text{if } i = j \\ -1 & \text{if } i \neq j \text{ and } v_i \text{ connects } v_j \\ 0 & \end{cases}$$

Laplacian matrix

Labelled graph	Degree matrix	Adjacency matrix	Laplacian matrix
	$\begin{pmatrix} 2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$	$\begin{pmatrix} 0 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \end{pmatrix}$	$\begin{pmatrix} 2 & -1 & 0 & 0 & -1 & 0 \\ -1 & 3 & -1 & 0 & -1 & 0 \\ 0 & -1 & 2 & -1 & 0 & 0 \\ 0 & 0 & -1 & 3 & -1 & -1 \\ -1 & -1 & 0 & -1 & 3 & 0 \\ 0 & 0 & 0 & -1 & 0 & 1 \end{pmatrix}$

$$L = D - W$$

$$P \in R^{m \times n}$$

$$n \leq m$$

$$\varepsilon(P) = \sum_{\{v_i, v_j\} \in E} \|\rho(v_i) - \rho(v_j)\|^2$$

$$L_{i,j} = \begin{cases} \deg(v_i) & \text{if } i = j \\ -1 & \text{if } i \neq j \text{ and } v_i \text{ connects } v_j \\ 0 & \end{cases}$$

($L = D - W$)

$$\varepsilon(P) \sim \text{tr}(P^T L P)$$

a	c	e
b	d	f

P^T

2	-1	-1
-1	2	-1
-1	-1	2

L

a	b
c	d
e	f

P

$2a-c-e$	$-a+2c-e$	$-a-c+2e$
$2b-d-f$	$-b+2d-f$	$-b-d+2f$

$$P^T L$$

a	b
c	d
e	f

$$P$$

2a	2c	2e
2b	2d	2f

$P^T L$

a	b
c	d
e	f

P

$2a^2+2c^2+2e^2$	$2ab+2cd+2ef$
$2ab+2cd+2ef$	$2b^2+2d^2+2f^2$

$$P^T L P$$

$$P \in R^{m \times n}$$

$$n \leq m$$

$$\varepsilon(P) = \sum_{\{v_i, v_j\} \in E} \|\rho(v_i) - \rho(v_j)\|^2$$

$$L_{i,j} = \begin{cases} \deg(v_i) & \text{if } i = j \\ -1 & \text{if } i \neq j \text{ and } v_i \text{ connects } v_j \\ 0 & \end{cases}$$

$$(L = D - W)$$



$$\varepsilon(P) \sim \text{tr}(P^T L P)$$

$$P \in R^{m \times n}$$

$$n \leq m$$

$$\varepsilon(P) = \sum_{\{v_i, v_j\} \in E} \|\rho(v_i) - \rho(v_j)\|^2$$

$$L_{i,j} = \begin{cases} \deg(v_i) & \text{if } i = j \\ -1 & \text{if } i \neq j \text{ and } v_i \text{ connects } v_j \\ 0 & \end{cases}$$

($L = D - W$)



$$\varepsilon(P) \sim \text{tr}(P^T L P)$$

$$P^T P = I$$

To satisfy $P^T P = I$:

$$\varepsilon(P) = \lambda_1 + \cdots \lambda_n$$

Kamada Kawai (KK)

$$\varepsilon = \sum_{i < j \leq |V|} k_{ij} (|p_i - p_j| - l_{ij})^2$$

$$\varepsilon = \sum_{i < j \leq |V|} k_{ij} \left(|p_i - p_j| - l_{ij} \right)^2$$

$$E = \sum_{i=1}^{n-1} \sum_{j=i+1}^n \frac{1}{2} k_{i,j} \left((x_i - x_j)^2 + (y_i - y_j)^2 + l_{i,j}^2 - 2l_{i,j} \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2} \right)$$

$$k_{i,j} = K/d_{i,j}^2$$

$$\varepsilon = \sum_{i < j \leq |V|} k_{ij} (|p_i - p_j| - l_{ij})^2$$

$$E = \sum_{i=1}^{n-1} \sum_{j=i+1}^n \frac{1}{2} k_{i,j} \left((x_i - x_j)^2 + (y_i - y_j)^2 + l_{i,j}^2 - 2l_{i,j} \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2} \right)$$

$$\uparrow \Delta_m = \sqrt{\left(\frac{\partial E}{\partial x_m}\right)^2 + \left(\frac{\partial E}{\partial y_m}\right)^2}.$$

$$\varepsilon = \sum_{i < j \leq |V|} k_{ij} (|p_i - p_j| - l_{ij})^2$$

$$E = \sum_{i=1}^{n-1} \sum_{j=i+1}^n \frac{1}{2} k_{i,j} \left((x_i - x_j)^2 + (y_i - y_j)^2 + l_{i,j}^2 - 2l_{i,j} \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2} \right)$$

$$\uparrow \Delta_m = \sqrt{\left(\frac{\partial E}{\partial x_m} \right)^2 + \left(\frac{\partial E}{\partial y_m} \right)^2}$$

https://networkx.org/documentation/stable/_modules/networkx/drawing/layout.html#kamada_kawai_layout

compute $d_{i,j}$ for $1 \leq i \neq j \leq n$;

compute $l_{i,j}$ for $1 \leq i \neq j \leq n$;

compute $k_{i,j}$ for $1 \leq i \neq j \leq n$;

initialize $p_1, p_2, \dots, p_n$;

while ($\max_i \Delta_i > \epsilon$)

 let p_m be the particle satisfying $\Delta_m = \max_i \Delta_i$;

while ($\Delta_m > \epsilon$)

 compute δx and δy by solving the following system of equations:

$$\frac{\partial^2 E}{\partial x_m^2}(x_m^{(t)}, y_m^{(t)})\delta x + \frac{\partial^2 E}{\partial x_m \partial y_m}(x_m^{(t)}, y_m^{(t)})\delta y = -\frac{\partial E}{\partial x_m}(x_m^{(t)}, y_m^{(t)});$$

$$\frac{\partial^2 E}{\partial y_m \partial x_m}(x_m^{(t)}, y_m^{(t)})\delta x + \frac{\partial^2 E}{\partial y_m^2}(x_m^{(t)}, y_m^{(t)})\delta y = -\frac{\partial E}{\partial y_m}(x_m^{(t)}, y_m^{(t)})$$

$$x_m := x_m + \delta x;$$

$$y_m := y_m + \delta y;$$

Fruchterman – Reingold (FR)

- Remember simulated annealing?
- In FR,
 - Each node has both attractive (F_a) and repulsive (F_r) forces.
 - Same as atoms that are electrified.
 - To stop them from moving, we lower the temperature.

Fruchterman – Reingold (FR)

- Advantages:
 - Good for huge data.
 - The distance between nodes are more balanced.

Pseudo-codes

- Initialize t as temperature.
- While True:
 - Calculate each vertex's force F .
 - Calculate new positions of each vertex with the new F .
 - Decide whether to apply moving vertex i using probability P .
 - If $P = 0$, stop.

III. Data-driven layout

- Based on data to determine position.
- Useful fields:
 - Coordinates
 - Country names/cities/places/...
 - Discrete values
- Too many overlapped points => Pick randomly.



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